Industrial Machine Failure Prediction for Predictive Maintenance

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1. Abstract

This project develops a machine-learning system for predicting industrial machine failure from process and sensor variables. The workflow covers exploratory data analysis, preprocessing, class-imbalance handling, model comparison, evaluation, feature importance and a maintenance-risk alert interface.

2. Objectives

Predict machine failure early; compare Logistic Regression, Decision Tree, Random Forest and SVM; evaluate precision, recall, F1-score and ROC-AUC; demonstrate an auxiliary regression workflow; and provide a practical prediction interface.

3. Dataset

The project follows the AI4I 2020 Predictive Maintenance Dataset structure published by the UCI Machine Learning Repository. The working file included in this project is a generated AI4I-style dataset based on the published structure and documented failure rules. The official UCI dataset contains 10,000 instances and is described as a synthetic dataset reflecting realistic predictive-maintenance conditions.

4. Methodology

Data validation → identifier and leakage removal → numerical imputation/scaling → categorical encoding → stratified train/test split → class-weighted model training → evaluation → best-model selection → risk prediction.

5. Classification

The target is Machine failure (0/1). Failure-mode labels TWF, HDF, PWF, OSF and RNF are not used as predictive inputs because they are derived from failure logic and would introduce target leakage.

6. Class Imbalance

Failure cases are much less frequent than normal cases. Class-weighted classifiers are used and recall is emphasized because missing a genuine failure may be more costly than generating an additional inspection alert.

7. Regression

An auxiliary regression experiment estimates continuous Tool wear [min] from the remaining operating variables. MAE, RMSE and R^2 are used to evaluate the regression models.

8. Deployment

A Streamlit prototype accepts machine type, temperatures, rotational speed, torque and tool wear and returns a failure probability with low-, medium- or high-risk maintenance guidance.

9. Limitations

The AI4I dataset is synthetic and the resulting model is a portfolio/academic prototype. Real industrial deployment would require validated plant-specific data, sensor calibration, drift monitoring, domain expertise and safety approval.

10. Future Scope

Real-time IoT integration, time-series modelling, Remaining Useful Life estimation, SHAP-based explanations, model monitoring, database integration and cloud deployment.

11. Classification Results

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Random Forest	0.886	0.662	0.882	0.756	0.949
Decision Tree	0.873	0.627	0.900	0.739	0.938
SVM	0.823	0.533	0.910	0.672	0.926
Logistic Regression	0.606	0.288	0.662	0.401	0.677

12. Regression Results

Model	MAE	RMSE	R^2
Random Forest Regressor	57.30	68.07	0.127
Linear Regression	60.87	70.39	0.066

13. Conclusion

The completed workflow demonstrates an end-to-end applied machine-learning project: data preparation, EDA, leakage-aware feature selection, imbalance-aware classification, model comparison, regression, evaluation and a deployable prediction prototype. Random Forest provides the strongest F1 and ROC-AUC among the tested classifiers on the included working dataset.